CVE Analysis Report

Report Title:
Analysis of CVE-2026-62817:

Authors/Group Members:
Blaise Billones
Arielle Blessy Reverente
Shayde Yamagata

Course/Instructor:
IT 381, Wolthius

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Executive Summary

CVE-2026-62817 with a CVSS score of 8.8 is a memory corruption vulnerability in Windows DNS Servers. Through exploitation the attacker can execute code in a vulnerable server from a neighboring network. Windows DNS Servers are commonly used in Active Directory domain controllers, as such a compromise could be extensive beyond just name resolution. Microsoft has released a fix on August 11, 2026, and affected servers should be patched as soon as possible.

Vulnerability Overview

- CVE Identifier: CVE-2026-62817 (Windows DNS Server Remote Code Execution Vulnerability)
- Affected Systems: Windows Server 2019, 2022, 2025
- Severity Metric (CVSS Score): 8.8/10 High; Microsoft rating of Critical
- CVSS Vector: CVSS:3.1/AV:A/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H
- Disclosure Date: August 11, 2026
- Type/Description: Out-of-bounds write (memory corruption) in Windows DNS. An unauthorized attacker can execute code on a neighboring network

Technical Analysis — What It Does and How It Works

CVE-2026-62817 is an out-of-bounds write vulnerability in the Windows DNS Server, a type of memory corruption flaw in which the service writes data past the end of a box of set buffer space and overwrites the adjacent memory. Because that neighboring memory can hold values the program uses to decide what to run next, an attacker who can control the overflowing data can also redirect the program to execute their own code. The attacker would most likely trigger the flaw by sending specially crafted network traffic to the DNS service. According to the CVSS vector, the attacker must be on an adjacent network, but the attack is low in complexity and needs no authentication information or user interaction. Successful exploitation results in code execution on the DNS server, with high impact to confidentiality, integrity, and availability.

Business/Organizational Impact

Successful exploitation may let an attacker run code on a DNS server with high impact on confidentiality, integrity and availability. Attackers may then:

- Disrupt name resolution so that people cant reach internal app,email,or file shares
- Steal or change data on the compromise server
- Move thru the network from a trusted infrastructure server
- Cause serious damage if the server is a domain controller that also hosts DNS

Remediation and Mitigation — How to Fix It

- Short Term do right away
    - Install the update from Microsoft –as it made a fix for this bug. Put it on every Windows Server that runs DNS (the “phone book” of the internet). Do the most important serverfirst which are the domain controller.They control who is allow to log in.
    - Make sure the update worked. Check the server version number . It should be the same as or newer than these:
        - Windows Server 2025: 10.0.26100.33296
        - Windows Server 2022: 10.0.20348.5499
        - Windows Server 2019: 10.0.17763.9115
- Medium Term (for next few weeks)
    - Keep DNS servers in a safe spot on the network. This attack only work if the hacker is on the same network as the server so keeping stranger device away from it helps Think of it like keeping the vault in a locked back room, not in the lobby.
    - Use firewalls or VLANs. These split one big network into smaller sections like lock rooms in a building.Then a random laptop or phone cant get close to the DNS servers.
    - Watch the login records. If a hacker did get in strange logins might show it, like a security camera catching something odd.

- Long Term (good habits to keep)
    - Make a list of which servers run DNS. You can't fix what you forgot you had.
    - Update on a regular schedule. Do it every month, and update faster when a bug is serious like this one.
    - Only run DNS on servers that need it. Fewer servers running it means fewer things a hacker can attack.

Verification and Lessons Learned

After installing the security update we need to verify if the vulnerability has been fixed. The easiest way to do this is by checking the Windows Server Build number and running another vulnerability scan using tools like Nessus or OpenVAS. We should also test if the DNS server is still working properly and check the security logs for any activities that are unusual. We have also learned that keeping servers updated is extremely important especially for services like DNS that many computers depend on. A one vulnerable server is all it takes to put the entire network at risk. Firewalls, VLAN's, and monitoring activities can help prevent attacks. If a server was already compromised, we should also investigate it before allowing it back on the network. The most important lesson learned here is that we should never wait for an attack before fixing the security problems.

Supporting Materials

Microsoft Security Response Center (MSRC)
https://msrc.microsoft.com/update-guide/vulnerability/CVE-2026-62817
This is the official page in Microsoft that talks about the vulnerability and available security updates.

National Vulnerability Database (NVD)
https://nvd.nist.gov/vuln/detail/cve-2026-62817
This website explains the vulnerability, the CVSS score, and how serious the security risk is.

OpenCVE
https://app.opencve.io/cve/CVE-2026-62817
This website provides information about affected Windows Server versions, attack methods, and recommended fixes.

Microsoft Windows Server 2019 Security Update
https://support.microsoft.com/en-us/servicing/os/windows-10/2026/08/kb5120238-windows-10-1809-security-update
This page provides information about the August 2026 security update for Windows Server 2019.

Citations

NVD entry: https://nvd.nist.gov/vuln/detail/cve-2026-62817

MSRC entry: https://msrc.microsoft.com/update-guide/vulnerability/CVE-2026-62817

SentinelOne Vulnerability Database: https://www.sentinelone.com/vulnerability-database/cve-2026-62817/Action1: https://www.action1.com/vulnerabilities/cve-2026-62817/

Appaloosa Scout (fixed OS builds): https://scout.appaloosa.io/cves/CVE-2026-62817

OpenCVE: https://app.opencve.io/cve/CVE-2026-62817